Assignment 1

- **Released date:** 2024/06/05.
- **Due:** 2024/06/13.
- Late submission is **NOT** acceptable.
- Please submit your assignment as a PDF file titled "student ID + HW1".

Question 1. In each of the following systems, interpret each equation as a line in the plane. For each system, graph the lines and determine geometrically the number of solutions.

(a)

$$\begin{aligned} x_1 + x_2 &= 4 \\ x_1 - x_2 &= 2 \end{aligned}$$

(b)

$$\begin{aligned} x_1 + 2x_2 &= 4 \\ -2x_1 - 4x_2 &= 4 \end{aligned}$$

(c)

$$\begin{aligned} 2x_1 - x_2 &= 3 \\ -4x_1 + 2x_2 &= -6 \end{aligned}$$

(d)

$$\begin{aligned} x_1 + x_2 &= 1 \\ x_1 - x_2 &= 1 \\ -x_1 + 3x_2 &= 3 \end{aligned}$$

Solution

(a) One solution. The two lines intersect at the point (3, 1).

(b) No solution. The lines are parallel.

(c) Infinitely many solutions. Both equations represent the same line.

(d) No solution. Each pair of lines intersect in a point; however, there is no point that is on all three lines.

Question 2. Write an augmented matrix for each of the systems in Question 1.

Solution

(a) $\left(\begin{array}{cc|c} 1 & 1 & 4 \\ 1 & -1 & 2 \end{array}\right)$; (b) $\left(\begin{array}{cc|c} 1 & 2 & 4 \\ -2 & -4 & 4 \end{array}\right)$; (c) $\left(\begin{array}{cc|c} 2 & -1 & 3 \\ -4 & 2 & -6 \end{array}\right)$;

(d) $\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 3 & 3 \end{array}\right)$

Question 3. The two systems

$$\begin{aligned} 2x_1 + x_2 &= 3 \\ 4x_1 + 3x_2 &= 5 \end{aligned} \quad \text{and} \quad \begin{aligned} 2x_1 + x_2 &= -1 \\ 4x_1 + 3x_2 &= 1 \end{aligned}$$

have the same coefficient matrix but different righthand sides. Solve both systems simultaneously by eliminating the first entry in the second row of the augmented matrix

$$\left(\begin{array}{cc|cc} 2 & 1 & 3 & -1 \\ 4 & 3 & 5 & 1 \end{array}\right)$$

and then performing back substitutions for each of the columns corresponding to the right-hand sides.

Solution

(a) (2, -1); (b) (-2, 3)

Question 4. Given a system of the form

$$\begin{aligned} -m_1 x_1 + x_2 &= b_1 \\ -m_2 x_1 + x_2 &= b_2 \end{aligned}$$

where m_1, m_2, b_1, and b_2 are constants:

(a) Show that the system will have a unique solution if $m_1 \neq m_2$.

(b) Show that if $m_1 = m_2$, then the system will be consistent only if $b_1 = b_2$.

(c) Give a geometric interpretation of parts (a) and (b).

Solution

(a) When $m_1 = 0, m_2 \neq 0$, the augmented matrix for the system is

$$\left(\begin{array}{cc|c} 0 & 1 & b_1 \\ -m_2 & 1 & b_2 \end{array}\right)$$

By substitution, we could obtain a unique solution $\left(\frac{b_1-b_2}{m_2}, b_1\right)$.
When $m_2 = 0, m_1 \neq 0$, the augmented matrix for the system is

$$\left(\begin{array}{cc|c} -m_1 & 1 & b_1 \\ 0 & 1 & b_2 \end{array}\right)$$

By substitution, we could obtain a unique solution $\left(\frac{b_2-b_1}{m_1}, b_2\right)$.
When $m_1 \neq 0, m_2 \neq 0$, the augmented matrix for the system is

$$\left(\begin{array}{cc|c} -m_1 & 1 & b_1 \\ -m_2 & 1 & b_2 \end{array}\right) \overset{R_2 \to R_2 - \frac{m_2}{m_1} R_1}{\longrightarrow} \left(\begin{array}{cc|c} -m_1 & 1 & b_1 \\ 0 & 1 - \frac{m_2}{m_1} & b_2 - \frac{b_1 m_2}{m_1} \end{array}\right)$$

By substitution, we could obtain a unique solution $\left(\frac{b_2-b_1}{m_1-m_2}, \frac{m_1 b_2 - b_1 m_2}{m_1-m_2}\right)$.
(b) When $m_1 = m_2$, the augmented matrix for the system is

$$\left(\begin{array}{cc|c} -m_1 & 1 & b_1 \\ -m_2 & 1 & b_2 \end{array}\right) \overset{R_2 \to R_2 - R_1}{\longrightarrow} \left(\begin{array}{cc|c} -m_1 & 1 & b_1 \\ 0 & 0 & b_2 - b_1 \end{array}\right)$$

So only if $b_2 - b_1 = 0$, that is $b_1 = b_2$, the system will be consistent.

(c) For part (a):
If $m_1 \neq m_2$, the two lines will intersect at one point, that is a unique solution.
For part(b) :
If $b_1 \neq b_2$, the two lines will be parallel, that is no solution.
Only if $b_1 = b_2$, both equations will represent the same line, that is infinite solutions.

Question 5. Consider a system of the form

$$a_{11}x_1 + a_{12}x_2 = 0$$
$$a_{21}x_1 + a_{22}x_2 = 0$$

where a_{11}, a_{12}, a_{21}, and a_{22} are constants. Explain why a system of this form must be consistent.

Solution

When $a_{11} \neq 0$, the augmented matrix for the system is

$$\left(\begin{array}{cc|c} a_{11} & a_{12} & 0 \\ a_{21} & a_{22} & 0 \end{array}\right) \overset{R_2 \to R_2 - \frac{a_{21}}{a_{11}} R_1}{\longrightarrow} \left(\begin{array}{cc|c} a_{11} & a_{12} & 0 \\ 0 & \frac{a_{22}a_{11} - a_{12}a_{21}}{a_{11}} & 0 \end{array}\right)$$

If $a_{22}a_{11} - a_{12}a_{21} = 0$, x_2 will be free variable, then the system is consistent.
If $a_{22}a_{11}a_{12}a_{21} \neq 0$, the system will have a unique solution (0, 0).
Obviously, the results above can be expanded to the situation that at least one of the four constants $\neq 0$.
And when all four constants $= 0$, the system will have infinite solutions.
So a system of this form must be consistent.

Question 6. Which of the matrices that follow are in the row echelon form? Which are in the reduced row echelon form?

(a)
$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 2 \end{pmatrix}$$

(b)
$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(c)
$$\begin{pmatrix} 1 & 3 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

(d)
$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

(e)
$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{pmatrix}$$

(f)

$$\begin{pmatrix} 1 & 4 & 6 \\ 0 & 0 & 1 \\ 0 & 1 & 3 \end{pmatrix}$$

(g)

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & 2 & 4 \\ 0 & 0 & 1 & 3 & 6 \end{pmatrix}$$

(h)

$$\begin{pmatrix} 0 & 1 & 3 & 4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Solution

Row echelon form: (a), (c), (d), (e), (g), (h).
Reduced row echelon form: (c), (d), (g).

Question 7. Use Gauss–Jordan reduction to solve each of the following systems.

(a)

$$x_1 + x_2 = -1$$
$$4x_1 - 3x_2 = 3$$

(b)

$$x_1 + 3x_2 + x_3 + x_4 = 3$$
$$2x_1 - 2x_2 + x_3 + 2x_4 = 8$$
$$3x_1 + x_2 + 2x_3 - x_4 = -1$$

(c)

$$x_1 + x_2 + x_3 = 0$$
$$x_1 - x_2 - x_3 = 0$$

(d)

$$x_1 + x_2 + x_3 + x_4 = 0$$
$$2x_1 + x_2 - x_3 + 3x_4 = 0$$
$$x_1 - 2x_2 + x_3 + x_4 = 0$$

Solution

(a) $(0, -1)$;
(b) $\left\{ \left(\frac{3}{4} - \frac{5}{8}\alpha, -\frac{1}{4} - \frac{1}{8}\alpha, \alpha, 3 \right) \mid \alpha \text{ is real } \right\}$;

(c) $\{0, \alpha, -\alpha\}$;
(d) $\left\{\alpha\left(-\frac{4}{3}, 0, \frac{1}{3}, 1\right)\right\}$

Question 8. Consider a linear system whose augmented matrix is of the form

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ -1 & 4 & 3 & 2 \\ 2 & -2 & a & 3 \end{array}\right)$$

For what values of a will the system have a unique solution?

Solution

The REF can be obtained from the given augmented matrix by elementary row operations as

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 6 & 4 & 3 \\ 0 & 0 & a+2 & 4 \end{array}\right]$$

to make the system have an unique solution, $a + 2 \neq 0 \Rightarrow a \neq -2$.

Question 9. Consider a linear system whose augmented matrix is of the form

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 1 & 3 & a & b \end{array}\right]$$

(a) For what values of a and b will the system have infinitely many solutions?

(b) For what values of a and b will the system be inconsistent?

Solution

(a) A row-echelon form for the system is

$$\left[\begin{array}{cccc} 1 & 1 & 3 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & a-5 & b-4 \end{array}\right]$$

There will be infinitely many solutions if there is a free variable, and this can only happen here is the the bottom row is $\left[\begin{array}{cccc} 0 & 0 & u & v \end{array}\right]$, where $u = 0$ and $v = 0$. Thus, $a = 5$ and $b = 4$.
(b) This can only happen here is the the bottom row is $\left[\begin{array}{cccc} 0 & 0 & u & v \end{array}\right]$, where $u = 0$ and $v \neq 0$. Thus, $a = 5$ and $b \neq 4$.

Question 10. Given a homogeneous system of linear equations, if the system is overdetermined, what are the possibilities as to the number of solutions? Explain.

Solution

Firstly, the homogeneous system is always consistent (because there is a trivial, all-zero solution).
If this system is overdetermined, then there are two cases, depending on the number of linearly dependent equations: either there is just the trivial solution, or there is the trivial solution plus an infinite set of other solutions.

Question 11. If

$$A = \begin{pmatrix} 3 & 1 & 4 \\ -2 & 0 & 1 \\ 1 & 2 & 2 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 & 0 & 2 \\ -3 & 1 & 1 \\ 2 & -4 & 1 \end{pmatrix}$$

compute

(a) $2A$ (b) $A + B$
(c) $2A - 3B$ (d) AB

Solution

(a) $\begin{pmatrix} 6 & 2 & 8 \\ -4 & 0 & 2 \\ 2 & 4 & 4 \end{pmatrix}$

(b) $\begin{pmatrix} 4 & 1 & 6 \\ -5 & 1 & 2 \\ 3 & -2 & 3 \end{pmatrix}$

(c) $\begin{pmatrix} 3 & 2 & 2 \\ 5 & -3 & -1 \\ -4 & 16 & 1 \end{pmatrix}$

(d) $\begin{pmatrix} 8 & -15 & 11 \\ 0 & -4 & -3 \\ -1 & -6 & 6 \end{pmatrix}$

Question 12. For each of the pairs of matrices that follow, determine whether it is possible to multiply the first matrix times the second. If it is possible, perform the multiplication.

(a) $\begin{pmatrix} 3 & 5 & 1 \\ -2 & 0 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 3 \\ 4 & 1 \end{pmatrix}$

(b) $\begin{pmatrix} 4 & -2 \\ 6 & -4 \\ 8 & -6 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$

(c) $\begin{bmatrix} 1 & 4 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 1 & 1 \\ 4 & 5 \end{bmatrix}$

(d) $\begin{bmatrix} 4 & 6 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & 5 \\ 4 & 1 & 6 \end{bmatrix}$

(e) $\begin{bmatrix} 4 & 6 & 1 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & 5 \\ 4 & 1 & 6 \end{bmatrix}$

(f) $\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \begin{bmatrix} 3 & 2 & 4 & 5 \end{bmatrix}$

Solution

(a) $\begin{pmatrix} 15 & 19 \\ 4 & 0 \end{pmatrix}$; (c) $\begin{pmatrix} 19 & 21 \\ 17 & 21 \\ 8 & 10 \end{pmatrix}$; (d) $\begin{pmatrix} 36 & 10 & 56 \\ 10 & 3 & 16 \end{pmatrix}$; (f) $\begin{pmatrix} 6 & 4 & 8 & 10 \\ -3 & -2 & -4 & -5 \\ 9 & 6 & 12 & 15 \end{pmatrix}$

(b) and (e) are not possible.

Question 13. Let A be a 5×3 matrix. If

$$\mathbf{b} = \mathbf{a}_1 + \mathbf{a}_2 = \mathbf{a}_2 + \mathbf{a}_3$$

then what can you conclude about the number of solutions of the linear system $A\mathbf{x} = \mathbf{b}$? Explain.

Solution

Since $\mathbf{b} = \mathbf{a}_1 + \mathbf{a}_2 = \mathbf{a}_2 + \mathbf{a}_3$, $\mathbf{a}_1 = \mathbf{a}_3$. So $A\mathbf{x} = \mathbf{b}$ can be represented by

$$\mathbf{a}_1 x_1 + \mathbf{a}_2 x_2 + \mathbf{a}_3 x_3 = \mathbf{a}_1(x_1 + x_3) + \mathbf{a}_2 x_2 = \mathbf{b} = \mathbf{a}_1 + \mathbf{a}_2.$$

Therefore, one of x_1 or x_3 will be a free variable, this linear system has **infinite solutions**.

Question 14. Let A be a 3×4 matrix. If

$$\mathbf{b} = \mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3 + \mathbf{a}_4$$

then what can you conclude about the number of solutions to the linear system $A\mathbf{x} = \mathbf{b}$? Explain.

Solution

Since $\mathbf{b} = \mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3 + \mathbf{a}_4$, it follows that $\mathbf{x} = (1, 1, 1, 1)^T$ is a solution to the equation $A\mathbf{x} = \mathbf{b}$.

And please note that A be a 3×4 matrix, the equation $A\mathbf{x} = \mathbf{b}$ is a under-determined linear system.

Since the matrix contains 3 rows there can be 3 nonzero rows and consequently 3 lead variables at most after we compute the row echelon form of the matrix.

So at least one of the variables is a free variable.

Therefore, the system has **infinitely many solutions**.

Question 15. Let $A\mathbf{x} = \mathbf{b}$ be a linear system whose augmented matrix $(A \mid \mathbf{b})$ has reduced row echelon form

$$\left(\begin{array}{ccccc|c} 1 & 2 & 0 & 3 & 1 & -2 \\ 0 & 0 & 1 & 2 & 4 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

(a) Find all solutions to the system.

(b) If

$$\mathbf{a}_1 = \begin{pmatrix} 1 \\ 1 \\ 3 \\ 4 \end{pmatrix} \quad \text{and} \quad \mathbf{a}_3 = \begin{pmatrix} 2 \\ -1 \\ 1 \\ 3 \end{pmatrix}$$

determine $\mathbf{b}$.

Solution

(a)

$$\begin{cases} x_1 + 2x_2 + 3x_4 + x_5 = -2 \\ x_3 + 2x_4 + 4x_5 = 5 \end{cases}$$

So $x_1 = -2 - 2x_2 - 3x_4 - x_5$, $x_3 = 5 - 2x_4 - 4x_5$, where x_2, x_4, x_5 are free variables.

(b) Since x_2, x_4, x_5 are free variables, we can assume $x_2 = x_4 = x_5 = 0$, then $x_1 = -2, x_3 = 5$. So

$$\mathbf{b} = \mathbf{a}_1 x_1 + \mathbf{a}_3 x_3 = \begin{pmatrix} 8 \\ -7 \\ -1 \\ 7 \end{pmatrix}$$

Question 16. Let A be a 2×2 matrix with $a_{11} \neq 0$ and let $\alpha = a_{21}/a_{11}$. Show that A can be factored into a product of the form

$$\begin{pmatrix} 1 & 0 \\ \alpha & 1 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ 0 & b \end{pmatrix}$$

What is the value of b?

Solution

When computing the product, the result is:

$$\begin{pmatrix} a_{11} & a_{12} \\ \alpha \cdot a_{11} & \alpha \cdot a_{12} + b \end{pmatrix}.$$

We already know $\alpha = a_{21}/a_{11}$, so $b = a_{22} - \frac{a_{12}a_{21}}{a_{11}}$.